CD74 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CD74 gene fusions from the msk_impact_50k_2026 study, covering 67 fusion events. 39/57 fusions with known frame status (68.4%) were in-frame. 59/66 fusions (89.4%) retained the CLIP, MHC2 interacting domain. Domain retention was tested with Fisher's exact test (p=0.270083, not statistically significant at alpha=0.05). No literature benchmark is configured for CD74.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 67 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ROS1, observed in 55 events.

Domain retention

CLIP, MHC2 interacting domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 36/39 (92.3%) of in-frame fusions retained the domain, p=0.270083 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Domain-disruption analysis was skipped: CD74 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

72% of 46 fusions with this gene as the 5' partner join at exon 6, against 14% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 53 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 148 aa (exon 5; label-permutation p=0.43956). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 12 aa from the nearest configured domain boundary at 136 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=46) and "not_retained" (n=7). For Frame_status, retained: 29/41 (70.7%, 95% CI 55.5%-82.4%); not_retained: 3/7 (42.9%, 95% CI 15.8%-75.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 92.2609 vs 34.4286, p=0.129535 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for CD74 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (3 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: CD74 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: CD74 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (31 row(s) in window_scan, 13 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ROS1	55	0.8208955223880597	0.5000004342923104	None	None	0.5699877094051449	confidence_certainty, domain_retention, recurrence	0.6588731326615538	1
NRG1	9	0.13432835820895522	0.5000004342923104	None	None	0.5699877094051449	confidence_certainty, domain_retention, recurrence	0.3419959799635056	2
CAMK2A	2	0.029850746268656716	0.5000004342923104	None	None	0.5699877094051449	confidence_certainty, domain_retention, recurrence	0.29377554368336783	3
PDE6A	1	0.014925373134328358	0.5000004342923104	None	None	0.5699877094051449	confidence_certainty, domain_retention, recurrence	0.28688690992906246	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
147	1	45	0	7	None	1.0	-0.0
148	1	45	1	6	0.13333333333333333	0.24891146589259794	0.6039550975288366
209	35	11	6	1	0.5303030303030303	1.0	-0.0
210	36	10	6	1	0.6	1.0	-0.0
240	37	9	6	1	0.6851851851851852	1.0	-0.0
248	38	8	6	1	0.7916666666666666	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

29	17
3	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
COAD	[0, 1], [0, 0]
CUP	[1, 0], [0, 0]
LUAD	[27, 15], [3, 2]
LUAS	[1, 0], [0, 0]
LUSC	[0, 1], [0, 0]
MDLC	[0, 0], [0, 1]
UEC	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	CLIP, MHC2 interacting	53	46	0	7	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
1	46	54	0.8518518518518519

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CAMK2A	2	2	2	2
NRG1	9	9	9	7
PDE6A	1	1	1	1
ROS1	55	55	55	44

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
5	149783000	149784000	34	EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, EVT-P-0002912-T01-IM3-55, ... (34 total)
5	149782000	149783000	16	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004752-T01-IM5-39, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, ... (16 total)
5	149784000	149785000	3	EVT-P-0000569-T01-IM3-65, EVT-P-0033566-T02-IM6-13, EVT-P-0064684-T01-IM7-1

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
209	39	39	1332	False
273	9	9	357	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	6	33	46	0.717391304347826	0.13726415094339622	18.259526629729578
five_prime	7	12	46	0.2608695652173913	0.06481132075471698	4.574646112174917
three_prime	7	5	7	0.7142857142857143	0.14707547169811322	2.9534054252829125
five_prime	4	1	46	0.021739130434782608	0.10764150943396227	0.002310732830489092
three_prime	5	1	7	0.14285714285714285	0.11075471698113208	0.2515750814160777
three_prime	8	1	7	0.14285714285714285	0.052641509433962265	0.5014935592350415

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
9	209	200	41	12	35	11	0.5303030303030303	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (41 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
10	210	200	42	11	36	10	0.6	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
40	240	200	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
48	248	200	44	9	38	8	0.7916666666666666	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (44 total)
109	209	100	41	12	35	11	0.5303030303030303	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (41 total)
110	210	100	42	11	36	10	0.6	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
140	240	100	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
147	247	100	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
148	248	100	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
159	209	50	39	14	34	12	1.1333333333333333	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (39 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
160	210	50	40	13	35	11	1.2727272727272727	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (40 total)
184	209	25	39	14	34	12	1.1333333333333333	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (39 total)
185	210	25	40	13	35	11	1.2727272727272727	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (40 total)
190	240	50	41	12	36	10	1.44	0.6499573253629032	0.18711515715379518	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (41 total)
198	248	50	42	11	37	9	1.6444444444444444	0.6256488062908825	0.20366937971157345	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
209	234	25	40	13	35	11	1.2727272727272727	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (40 total)
209	259	50	42	11	37	9	1.6444444444444444	0.6256488062908825	0.20366937971157345	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
210	310	100	12	41	11	35	1.8857142857142857	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0014506-T01-IM6-52, ... (12 total)
210	410	200	12	41	11	35	1.8857142857142857	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0014506-T01-IM6-52, ... (12 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
223	273	50	11	42	10	36	1.6666666666666667	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0019346-T01-IM6-40, ... (11 total)
240	290	50	11	42	10	36	1.6666666666666667	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0019346-T01-IM6-40, ... (11 total)
240	340	100	11	42	10	36	1.6666666666666667	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0019346-T01-IM6-40, ... (11 total)
240	440	200	11	42	10	36	1.6666666666666667	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0019346-T01-IM6-40, ... (11 total)
248	273	25	10	43	9	37	1.4594594594594594	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0025107-T01-IM6-44, ... (10 total)
248	298	50	10	43	9	37	1.4594594594594594	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0025107-T01-IM6-44, ... (10 total)
248	348	100	10	43	9	37	1.4594594594594594	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0025107-T01-IM6-44, ... (10 total)
248	448	200	10	43	9	37	1.4594594594594594	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0025107-T01-IM6-44, ... (10 total)
273	298	25	9	44	8	38	1.263157894736842	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0025107-T01-IM6-44, EVT-P-0027196-T01-IM6-71, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
273	323	50	9	44	8	38	1.263157894736842	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0025107-T01-IM6-44, EVT-P-0027196-T01-IM6-71, ... (9 total)
273	373	100	9	44	8	38	1.263157894736842	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0025107-T01-IM6-44, EVT-P-0027196-T01-IM6-71, ... (9 total)
273	473	200	9	44	8	38	1.263157894736842	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0025107-T01-IM6-44, EVT-P-0027196-T01-IM6-71, ... (9 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
198	248	50	42	11	37	9	1.6444444444444444	0.6256488062908825	0.20366937971157345	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
190	240	50	41	12	36	10	1.44	0.6499573253629032	0.18711515715379518	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (41 total)
184	209	25	39	14	34	12	1.1333333333333333	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (39 total)
185	210	25	40	13	35	11	1.2727272727272727	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (40 total)
248	273	25	10	43	9	37	1.4594594594594594	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0025107-T01-IM6-44, ... (10 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
273	298	25	9	44	8	38	1.263157894736842	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0025107-T01-IM6-44, EVT-P-0027196-T01-IM6-71, ... (9 total)
223	273	50	11	42	10	36	1.6666666666666667	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0019346-T01-IM6-40, ... (11 total)
109	209	100	41	12	35	11	0.5303030303030303	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (41 total)
110	210	100	42	11	36	10	0.6	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (42 total)
140	240	100	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
148	248	100	43	10	37	9	0.6851851851851852	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (43 total)
210	310	100	12	41	11	35	1.8857142857142857	1.0	-0.0	EVT-P-0002877-T01-IM3-53, EVT-P-0004747-T01-IM5-48, EVT-P-0004790-T01-IM5-56, EVT-P-0011180-T01-IM5-42, EVT-P-0014506-T01-IM6-52, ... (12 total)
48	248	200	44	9	38	8	0.7916666666666666	1.0	-0.0	EVT-P-0000569-T01-IM3-65, EVT-P-0000632-T01-IM3-68, EVT-P-0000840-T01-IM3-46, EVT-P-0001437-T01-IM3-45, EVT-P-0002707-T01-IM3-49, ... (44 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064684-T01-IM7-1	P-0064684-T01-IM7	P-0064684	Breast Cancer	MDLC	CAMK2A-CD74	CAMK2A	out-of-frame	three_prime	149784826	5	148	True	148	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		CAMK2A (NM_015981) - CD74 (NM_001025159) rearrangement: c.63-1328:CAMK2A_c.442-83:CD74dup | Protein Fusion: out of frame {CAMK2A:CD74}	NA
EVT-P-0035570-T01-IM6-2	P-0035570-T01-IM6	P-0035570	Endometrial Cancer	UEC	CAMK2A-CD74	CAMK2A	out-of-frame	three_prime	149783112	7	209	True	209	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		CAMK2A (NM_015981)-CD74 (NM_001025159) Rearrangement: c.985-2282:CAMK2A_c.626-237:CD74dup | Protein Fusion: out of frame {CAMK2A:CD74}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060522-T01-IM7-10	P-0060522-T01-IM7	P-0060522	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	NRG1	in-frame	five_prime	149783541	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149783541::chr8:g.32570123) | Protein Fusion: in frame {CD74:NRG1}	NA
EVT-P-0058867-T01-IM7-11	P-0058867-T01-IM7	P-0058867	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	NRG1	in-frame	five_prime	149783529	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149783529::chr8:g.32565439) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058935-T02-IM7-12	P-0058935-T02-IM7	P-0058935	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	NRG1	in-frame	five_prime	149783327	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) fusion: t(5;8)(q32;p12)(chr5:g.149783327::chr8:g.32574604) | Protein Fusion: in frame {CD74:NRG1}	NA
EVT-P-0033566-T02-IM6-13	P-0033566-T02-IM6	P-0033566	Colorectal Cancer	COAD	CD74-PDE6A	PDE6A	out-of-frame	five_prime	149784884	4	147	True	147	True	exact match	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - PDE6A (NM_000440) rearrangement: c.442-141:CD74_c.2135+2556:PDE6Adel | Protein Fusion: out of frame {CD74:PDE6A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004747-T04-IM7-14	P-0004747-T04-IM7	P-0004747	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149782562	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782562::chr6:g.117649107) | Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0026455-T01-IM6-15	P-0026455-T01-IM6	P-0026455	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783651	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783651::chr6:g.117646566) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039219-T01-IM6-16	P-0039219-T01-IM6	P-0039219	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783535	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783535::chr6:g.117646339) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0040577-T01-IM6-17	P-0040577-T01-IM6	P-0040577	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149783441	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 32-43) : t(5;6)(q32;q22.1)(chr5:g.149783441::chr6:g.117650159) | Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011025-T02-IM6-18	P-0011025-T02-IM6	P-0011025	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783756	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783756::chr6:g.117645889) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0040869-T01-IM6-19	P-0040869-T01-IM6	P-0040869	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783792	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783792::chr6:g.117647263) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000840-T03-IM6-20	P-0000840-T03-IM6	P-0000840	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783509	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783509::chr6:g.117646938) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0041118-T06-IM6-21	P-0041118-T06-IM6	P-0041118	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	unknown	five_prime	149782265	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647425) | Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044465-T01-IM6-22	P-0044465-T01-IM6	P-0044465	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783638	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783638::chr6:g.117646909) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0014506-T02-IM6-23	P-0014506-T02-IM6	P-0014506	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	unknown	five_prime	149782872	7	210	False	209	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623) | Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047502-T01-IM6-24	P-0047502-T01-IM6	P-0047502	Non-Small Cell Lung Cancer	LUAS	CD74-ROS1	ROS1	in-frame	five_prime	149782474	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782474::chr6:g.117646405) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0048671-T01-IM6-25	P-0048671-T01-IM6	P-0048671	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783869	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048671-T02-IM6-26	P-0048671-T02-IM6	P-0048671	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783869	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0052363-T02-IM7-27	P-0052363-T02-IM7	P-0052363	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783700	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783700::chr6:g.117647256) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054352-T01-IM6-28	P-0054352-T01-IM6	P-0054352	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149783536	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783536::chr6:g.117649394) | Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0049640-T01-IM6-29	P-0049640-T01-IM6	P-0049640	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149782214	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) | Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049640-T03-IM7-30	P-0049640-T03-IM7	P-0049640	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149782214	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) | Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0014506-T05-IM7-31	P-0014506-T05-IM7	P-0014506	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	unknown	five_prime	149782872	7	210	False	209	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623) | Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054497-T01-IM6-32	P-0054497-T01-IM6	P-0054497	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149783252	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783252::chr6:g.117659373) | Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0052363-T01-IM6-33	P-0052363-T01-IM6	P-0052363	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783692	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783692::chr6:g.117647256) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011025-T01-IM5-34	P-0011025-T01-IM5	P-0011025	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149783755	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 34-43) : t(5;6)(q32;q22.1)(chr5:g.149783755::chr6:g.117645889) | Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0054440-T02-IM6-35	P-0054440-T02-IM6	P-0054440	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	in-frame	five_prime	149782403	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782403::chr6:g.117646088) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024572-T01-IM6-36	P-0024572-T01-IM6	P-0024572	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	out-of-frame	five_prime	149783583	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion(CD74 exons 1-6 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783583::chr6:g.117649758) | Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0041118-T02-IM6-37	P-0041118-T02-IM6	P-0041118	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	ROS1	unknown	five_prime	149782265	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647424) | Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010260-T01-IM5-38	P-0010260-T01-IM5	P-0010260	Cancer of Unknown Primary	CUP	NRG1-CD74	NRG1	in-frame	five_prime	149783390	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused in frame with NRG1 exons 3-13): t(5;8)(q32;p12)(chr5:g.149783390::chr8:g.32546211) | Protein fusion: in frame (CD74-NRG1)	NA
EVT-P-0004752-T01-IM5-39	P-0004752-T01-IM5	P-0004752	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	NRG1	in-frame	five_prime	149782966	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement : t(5;8)(5q32;8p12)(chr8:g.32549502::chr5:g.149782966) | Protein fusion: in frame (CD74-NRG1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019346-T01-IM6-40	P-0019346-T01-IM6	P-0019346	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	NRG1	unknown	five_prime	149782781	7	240	False	240	True	exact match	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-7 fused with NRG1 exons 6-13) : t(5;8)(q32;p12)(chr5:g.149782781::chr8:g.32584466) | Protein Fusion: mid-exon {CD74:NRG1}	NA
EVT-P-0004752-T02-IM5-41	P-0004752-T02-IM5	P-0004752	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	NRG1	in-frame	five_prime	149782964	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused with NRG1 exons 6-13) : t(5;8)(5q32;8p12)(chr5:g.149782964::chr8:g.32549504) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011180-T01-IM5-42	P-0011180-T01-IM5	P-0011180	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	NRG1	unknown	five_prime	149782758	7	248	False	247	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149782758::chr8:g.32513214) | Protein fusion: mid-exon (CD74-NRG1)	NA
EVT-P-0004752-T03-IM6-43	P-0004752-T03-IM6	P-0004752	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	NRG1	in-frame	five_prime	149782964	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused with NRG1 exons 6-13) : t(5;8)(5q32;8p12)(chr5:g.149782964::chr8:g.32549504) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025107-T01-IM6-44	P-0025107-T01-IM6	P-0025107	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149782204	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782204::chr6:g.117647170) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0001437-T01-IM3-45	P-0001437-T01-IM3	P-0001437	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149783504	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			None | Protein fusion: out of frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000840-T01-IM3-46	P-0000840-T01-IM3	P-0000840	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783508	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			ROS1 (NM_002944) - CD74 (NM_001025158) Translocation: t(6;5)(q22.1;q32)(chr6:g.117646938::chr5:g.149783508) | Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0036970-T01-IM6-47	P-0036970-T01-IM6	P-0036970	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149783480	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q33.1;q22.1)(chr5:g.149783480::chr6:g.117650273) | Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004747-T01-IM5-48	P-0004747-T01-IM5	P-0004747	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149782561	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6) (q32;q22.1) (chr5:g.149782561::chr6:g.117649107) | Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-0002707-T01-IM3-49	P-0002707-T01-IM3	P-0002707	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	three_prime	149783660	7	209	True	209	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		NA | Protein fusion: in frame (ROS1-CD74)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002707-T03-IM5-50	P-0002707-T03-IM5	P-0002707	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783748	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.49783748::chr6:g.117646946) | Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0012525-T01-IM5-51	P-0012525-T01-IM5	P-0012525	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783203	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1(NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783203::chr6:g.117647385) | Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014506-T01-IM6-52	P-0014506-T01-IM6	P-0014506	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	unknown	five_prime	149782871	7	210	False	210	True	exact match	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149782871::chr6:g.117650623) | Protein fusion: mid-exon (CD74-ROS1)	NA
EVT-P-0002877-T01-IM3-53	P-0002877-T01-IM3	P-0002877	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149782435	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			NA | Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041118-T07-IM7-54	P-0041118-T07-IM7	P-0041118	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	three_prime	149782213	8	273	True	273	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149782213::chr6:g.117648241) | Protein Fusion: out of frame {ROS1:CD74}	NA
EVT-P-0002912-T01-IM3-55	P-0002912-T01-IM3	P-0002912	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	three_prime	149783419	7	209	True	209	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		NA | Protein fusion: in frame (ROS1-CD74)	NA
EVT-P-0004790-T01-IM5-56	P-0004790-T01-IM5	P-0004790	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149782207	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (exons 1-7 of CD74 fused to exons 34-43 of ROS1) : t(5;6)(q32;q22.1)(chr5:g.149782207::chr6:g.117646467) | Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015215-T01-IM6-57	P-0015215-T01-IM6	P-0015215	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149783057	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149783057::chr6:g.117650370) | Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-0016213-T01-IM6-58	P-0016213-T01-IM6	P-0016213	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149783418	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783418::chr6:g.117649786) | Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018222-T01-IM6-59	P-0018222-T01-IM6	P-0018222	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783915	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783915::chr6:g.117646209) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0018513-T01-IM6-60	P-0018513-T01-IM6	P-0018513	Non-Small Cell Lung Cancer	LUSC	ROS1-CD74	ROS1	unknown	five_prime	149783804	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 33 to 43): t(5;6)(q32;q22.1)(chr5:g.149783804::chr6:g.117647544) | Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007108-T01-IM5-61	P-0007108-T01-IM5	P-0007108	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783704	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1 to 6 fused with ROS1 exons 34 to 43) : t(5;6)(chr5q32;chr6q22.1)(chr5:g.149783704::chr6:g.117646132) | Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0021054-T01-IM6-62	P-0021054-T01-IM6	P-0021054	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783021	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783021::chr6:g.117647162) | Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021228-T01-IM6-63	P-0021228-T01-IM6	P-0021228	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783729	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43):t(5;6)(q32;q22.1)(chr5:g.149783729::chr6:g.117646295) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0000569-T01-IM3-65	P-0000569-T01-IM3	P-0000569	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	three_prime	149784206	7	209	True	209	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		NA | Protein fusion: out of frame (ROS1-CD74)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023617-T01-IM6-66	P-0023617-T01-IM6	P-0023617	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149782946	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_0012515) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782946::chr6:g.117646444) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0008134-T01-IM5-67	P-0008134-T01-IM5	P-0008134	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783121	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_0012515 9) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;6q22.1)(chr5:g.149783121::chr6:g.117646080) | Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000632-T01-IM3-68	P-0000632-T01-IM3	P-0000632	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783283	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			NA | Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0004336-T01-IM5-69	P-0004336-T01-IM5	P-0004336	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783071	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43) : t(5;6)(5q32;6q22.1)(chr5:g.149783071::chr6:g.117647249) | Protein fusion: in frame (CD74-ROS1)	NA

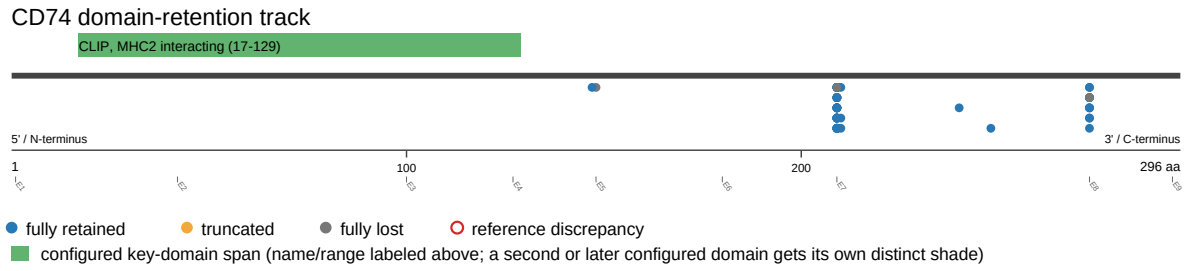
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026589-T02-IM6-70	P-0026589-T02-IM6	P-0026589	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149782990	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) Fusion (CD74 exon6 fused to ROS1 exon34) : t(5;6)(q32;q22.1)(chr5:g.149782990::chr6:g.117646032) | Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0027196-T01-IM6-71	P-0027196-T01-IM6	P-0027196	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	out-of-frame	five_prime	149782321	7	273	True	272	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149782321::chr6:g.117650377) | Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032722-T01-IM6-72	P-0032722-T01-IM6	P-0032722	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	unknown	five_prime	149782874	7	209	False	209	True	exact match	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782874::chr6:g.117645677) | Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-0010760-T01-IM5-73	P-0010760-T01-IM5	P-0010760	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	five_prime	149783763	6	209	True	208	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	CLIP, MHC2 interacting			CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;6q22.1)(chr5:g.149783763::chr6:g.117647234) | Protein fusion: in frame (CD74-ROS1)	NA

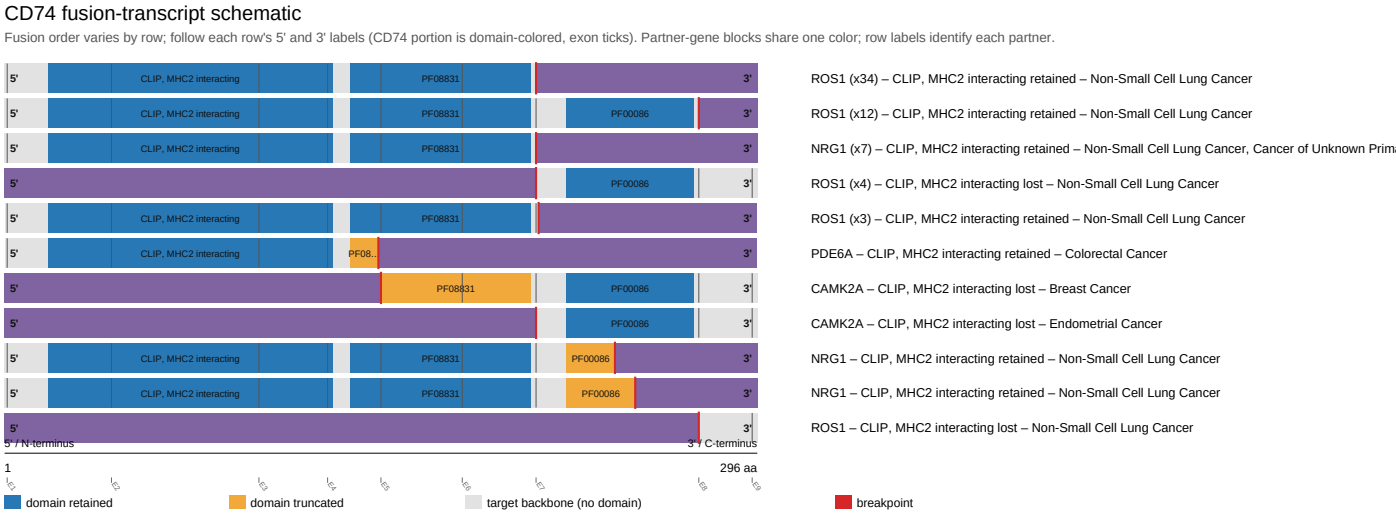
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065104-T01-IM7-74	P-0065104-T01-IM7	P-0065104	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	ROS1	in-frame	three_prime	149783694	7	209	True	209	True	exact match	lost	0.0	False		CLIP, MHC2 interacting		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149783694::chr6:g.117646386) | Protein Fusion: in frame {ROS1:CD74}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.